

YELLOWKNIFE, NT, Canada

WMO: 719360

Lat: 62.460N

Lon: 114.440W

Elev: 675

StdP: 14.34

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.9	-35.8	-47.5	0.3	-39.4	-44.4	0.4	-35.5	20.7	-3.0	18.5	-4.6	4.0	310	0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	78.0	61.0	75.0	59.6	72.2	58.4	63.0	73.2	61.5	71.4	60.1	69.6	9.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	76.9	66.3	57.4	72.2	65.3	55.6	67.5	64.6	28.9	73.2	27.8	71.5	26.8	69.7	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	18.2	16.3	DB	-43.2	83.4	3.6	2.8	-45.8	85.4	-47.9	87.1	-49.9	88.7	-52.5	90.7	(4)
(5)			WB	-43.3	65.5	3.5	2.0	-45.9	67.0	-47.9	68.1	-49.9	69.2	-52.4	70.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.5	-13.0	-7.8	2.4	22.2	41.5	57.1	62.9	57.9	46.3	30.6	9.8	-6.4
	DBStd	28.95	14.31	12.91	14.88	13.09	10.05	7.29	5.63	6.33	7.73	8.27	12.43	14.76
	HDD50	9918	1952	1619	1474	833	293	22	0	10	158	602	1206	1749
	HDD65	14491	2417	2039	1939	1283	727	251	106	232	560	1067	1656	2214
	CDD50	972	0	0	0	0	31	234	402	255	49	1	0	0
	CDD65	69	0	0	0	0	0	14	43	12	0	0	0	0
	CDH74	357	0	0	0	0	1	87	203	66	0	0	0	0
	CDH80	35	0	0	0	0	0	7	20	8	0	0	0	0
Wind	WSAvg	7.8	6.5	6.7	7.7	8.3	8.8	8.1	7.7	7.8	8.4	8.4	7.9	6.9
Precipitation	PrecAvg	11.3	0.6	0.6	0.5	0.4	0.7	1.0	1.4	1.7	1.3	1.5	1.0	0.6
	PrecMax	17.2	1.3	1.4	1.6	1.1	2.4	2.9	4.3	5.7	2.5	4.1	1.8	1.9
	PrecMin	7.0	0.0	0.1	0.1	0.0	0.0	0.1	0.2	0.2	0.2	0.4	0.3	0.2
	PrecStd	2.1	0.3	0.3	0.3	0.3	0.6	0.7	0.9	1.3	0.6	0.7	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	26.0	38.7	55.2	72.3	80.6	82.7	80.7	68.6	54.7	36.1	26.9
		MCWB	24.1	23.8	33.1	43.6	53.6	61.7	62.9	62.8	56.8	48.1	34.2	25.6
	2%	DB	18.3	19.4	33.6	48.6	66.4	76.4	78.8	75.1	63.7	47.7	32.7	23.1
		MCWB	17.3	17.6	29.6	39.8	50.8	59.0	61.8	60.1	54.6	43.4	31.3	22.1
	5%	DB	11.7	14.8	29.5	44.4	61.4	73.2	76.0	71.5	60.2	44.1	29.2	18.8
		MCWB	10.7	13.2	26.2	37.1	48.4	57.2	60.7	59.2	52.9	40.9	28.1	18.0
	10%	DB	6.7	11.0	25.0	40.5	57.1	69.9	73.2	68.1	57.2	41.0	26.1	13.9
		MCWB	5.7	9.6	22.2	34.7	45.9	56.0	59.4	58.1	51.4	38.5	25.0	13.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	23.8	33.7	44.0	54.9	63.3	65.7	64.7	58.8	49.0	34.6	26.0
		MCDB	26.0	26.0	38.1	54.1	69.1	76.4	77.1	74.5	65.9	54.3	36.3	26.9
	2%	WB	17.3	17.9	29.9	40.3	51.6	60.9	63.7	62.3	55.7	44.1	31.5	22.2
		MCDB	18.3	19.2	33.7	47.9	65.2	72.8	74.1	70.9	61.5	47.3	32.5	23.1
	5%	WB	10.8	13.2	26.3	37.5	49.1	59.0	62.2	60.6	53.7	41.1	28.2	17.9
		MCDB	11.7	14.7	29.6	43.8	60.4	70.2	72.3	68.8	59.4	44.0	29.2	18.7
	10%	WB	5.8	9.6	22.2	34.9	46.4	57.1	60.9	58.9	51.8	38.7	25.0	13.1
		MCDB	6.7	10.9	25.0	40.0	56.5	68.1	71.0	66.7	56.6	41.0	26.0	13.9
Mean Daily Temperature Range	5% DB	MDBR	12.6	15.2	19.4	19.6	17.5	16.1	14.5	12.9	11.0	8.3	11.6	12.7
		MCDBR	13.6	15.4	18.9	19.7	22.5	20.1	18.0	17.6	15.4	11.3	11.2	14.7
		MCWBR	13.4	14.6	15.8	13.3	12.3	9.2	7.3	7.6	8.1	8.3	10.8	14.4
	5% WB	MCDBR	13.7	15.3	19.0	18.5	21.6	18.3	16.0	15.3	14.6	10.8	11.3	14.5
		MCWBR	13.5	14.6	16.0	12.6	12.2	9.0	7.6	7.8	8.2	8.2	10.9	14.2
Clear-Sky Solar Irradiance	taub	0.193	0.243	0.270	0.305	0.326	0.345	0.369	0.339	0.310	0.275	0.192	0.156	
	taud	2.030	2.207	2.234	2.275	2.462	2.447	2.418	2.518	2.558	2.411	2.042	1.931	
	Ebn at Noon	159	233	268	280	284	279	268	268	257	220	155	111	
	Edh at Noon	10	19	28	33	30	32	32	26	21	16	10	6	
All-Sky Solar Radiation	RadAvg	82	331	885	1538	1874	1968	1736	1299	783	302	90	36	
	RadStd	10	28	66	84	116	168	91	62	66	34	13	3	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+2.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page